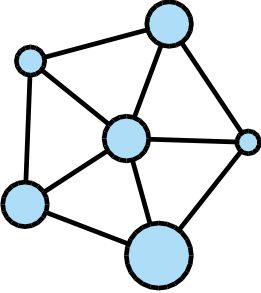
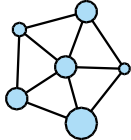
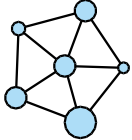


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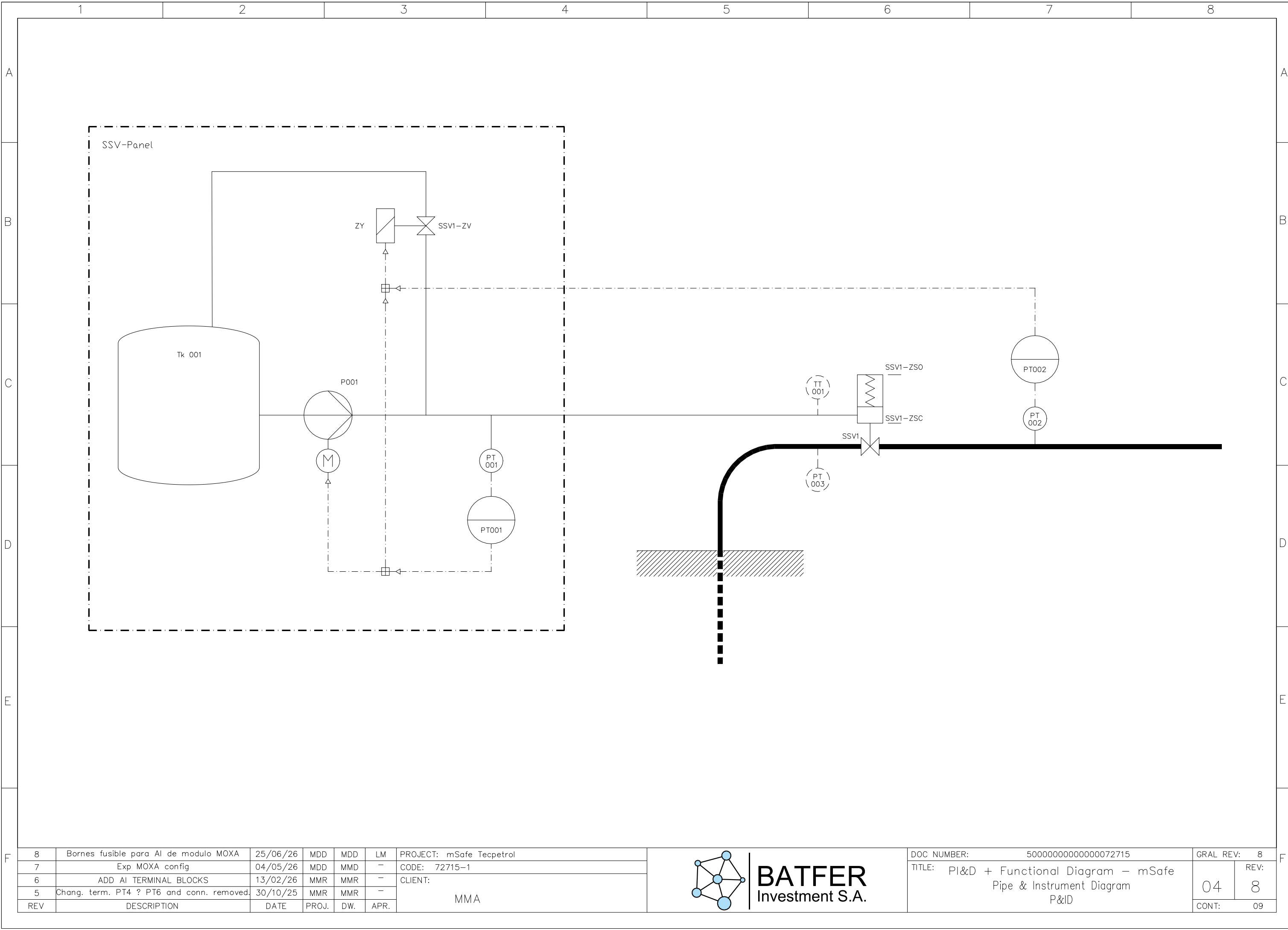
NOTE: cables not indicated will be black 0.75mm²/18 AWG

1	2	3	4	5	6	7	8																																				
A								A																																			
B	 BATFER Investment S.A.							B																																			
C								C																																			
D	Client: - MMA Project: - mSafe Tecpetrol Description Document: - 72715-1 Electrical Diagram mSafe Document Number: - 72715 ZPL Revision: - 8					APROBADO PARA PRODUCCION APROBADO: – FECHA: 25/06/2026  BATFER Investment S.A.			D																																		
E	Installation Place -							E																																			
F	<table><tr><td>8</td><td>Bornes fusible para AI de modulo MOXA</td><td>25/06/26</td><td>MDD</td><td>MDD</td><td>LM</td><td>PROJECT: mSafe Tecpetrol</td></tr><tr><td>7</td><td>Exp MOXA config</td><td>04/05/26</td><td>MDD</td><td>MMD</td><td>–</td><td>CODE: 72715–1</td></tr><tr><td>6</td><td>ADD AI TERMINAL BLOCKS</td><td>13/02/26</td><td>MMR</td><td>MMR</td><td>–</td><td>CLIENT:</td></tr><tr><td>5</td><td>Chang. term. PT4 ? PT6 and conn. removed.</td><td>30/10/25</td><td>MMR</td><td>MMR</td><td>–</td><td>MMA</td></tr><tr><td>REV</td><td>DESCRIPTION</td><td>DATE</td><td>PROJ.</td><td>DW.</td><td>APR.</td><td></td></tr></table>					8	Bornes fusible para AI de modulo MOXA	25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol	7	Exp MOXA config	04/05/26	MDD	MMD	–	CODE: 72715–1	6	ADD AI TERMINAL BLOCKS	13/02/26	MMR	MMR	–	CLIENT:	5	Chang. term. PT4 ? PT6 and conn. removed.	30/10/25	MMR	MMR	–	MMA	REV	DESCRIPTION	DATE	PROJ.	DW.	APR.		 BATFER Investment S.A.	DOC NUMBER: 50000000000000072715 TITLE: PI&D + Functional Diagram – mSafe COVER – GRAL REV: 8 REV: 8 CONT: 01	F
8	Bornes fusible para AI de modulo MOXA	25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol																																					
7	Exp MOXA config	04/05/26	MDD	MMD	–	CODE: 72715–1																																					
6	ADD AI TERMINAL BLOCKS	13/02/26	MMR	MMR	–	CLIENT:																																					
5	Chang. term. PT4 ? PT6 and conn. removed.	30/10/25	MMR	MMR	–	MMA																																					
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.																																						

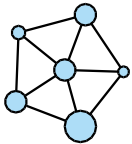
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1		2		3		4		5		6		7		8												
A	Pág.	Page Title		Revision								Additional Comments	Pág.	Page Title		Revision								Additional Comments		
				2	3	4	5	6	7	8	2					3	4	5	6	7	8					
		00	COVER				X	X	X	X					45	FREE										
01	INDEX				X	X	X	X			46	P&ID														
02	FREE										47	FREE														
03	FREE										48	FREE														
04	P&ID										49	FREE														
05	FREE										50	FREE														
06	FREE										51	FREE														
07	FREE										52	FREE														
08	FREE										53	FREE														
09	Material's List										54	FREE														
10	FREE										55	FREE														
11	Power Generation					X					56	FREE														
12	Battery Chargers and Power Circuits				X	X					57	FREE														
13	Control Voltage – Generation and Distribution	X			X	X		X			58	FREE														
14	Control Unit – Power Supply										59	FREE														
15	Intrinsic Safety Barrier – Power Supply										60	FREE														
16	Common Distribution for Signals	X			X						61	Digital Outputs –Main Board				X										
17	FREE										62	FREE														
18	FREE										63	FREE														
19	FREE										64	FREE														
20	FREE										65	FREE														
21	Electronic Board (21PCB01) – Main Board				X	X					66	FREE														
22	FREE										67	FREE														
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24	FREE										69	FREE														
25	FREE										70	FREE														
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27	FREE										72	FREE														
28	FREE										73	FREE														
29	FREE										74	FREE														
30	FREE										75	FREE														
31	Inner Analog Inputs – Main Board				X	X					76	FREE														
32	External Analog Inputs – Main Board				X	X					77	FREE														
33	External Analog Inputs – Analog Input Module	X			X			X			78	FREE														
34	FREE										79	FREE														
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36	FREE										81	Communications			X			X								
37	FREE										82	FREE														
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40	FREE										85	FREE														
41	FREE										86	FREE														
42	FREE										87	FREE														
43	External Digital Inputs – Main Board					X					88	FREE														
44	FREE										89	FREE														
F	8	Bornes fusible para AI de modulo MOXA		25/06/26		MDD	MDD	LM	PROJECT: mSafe Tecpetrol										DOC NUMBER: 50000000000000072715						GRAL REV: 8	
	7	Exp MOXA config		04/05/26		MDD	MMD	–	CODE: 72715–1										TITLE: PI&D + Functional Diagram – mSafe						REV: 8	
	6	ADD AI TERMINAL BLOCKS		13/02/26		MMR	MMR	–	CLIENT: MMA										INDEX						01	
	5	Chang. term. PT4 ? PT6 and conn. removed.		30/10/25		MMR	MMR	–																	8	
	REV	DESCRIPTION		DATE		PROJ.	DW.	APR.																	CONT: 04	

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8	Bornes fusible para AI de modulo MOXA	25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol
7	Exp MOXA config	04/05/26	MDD	MMD	-	CODE: 72715-1
6	ADD AI TERMINAL BLOCKS	13/02/26	MMR	MMR	-	CLIENT:
5	Chang. term. PT4 ? PT6 and conn. removed.	30/10/25	MMR	MMR	-	MMA
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.	



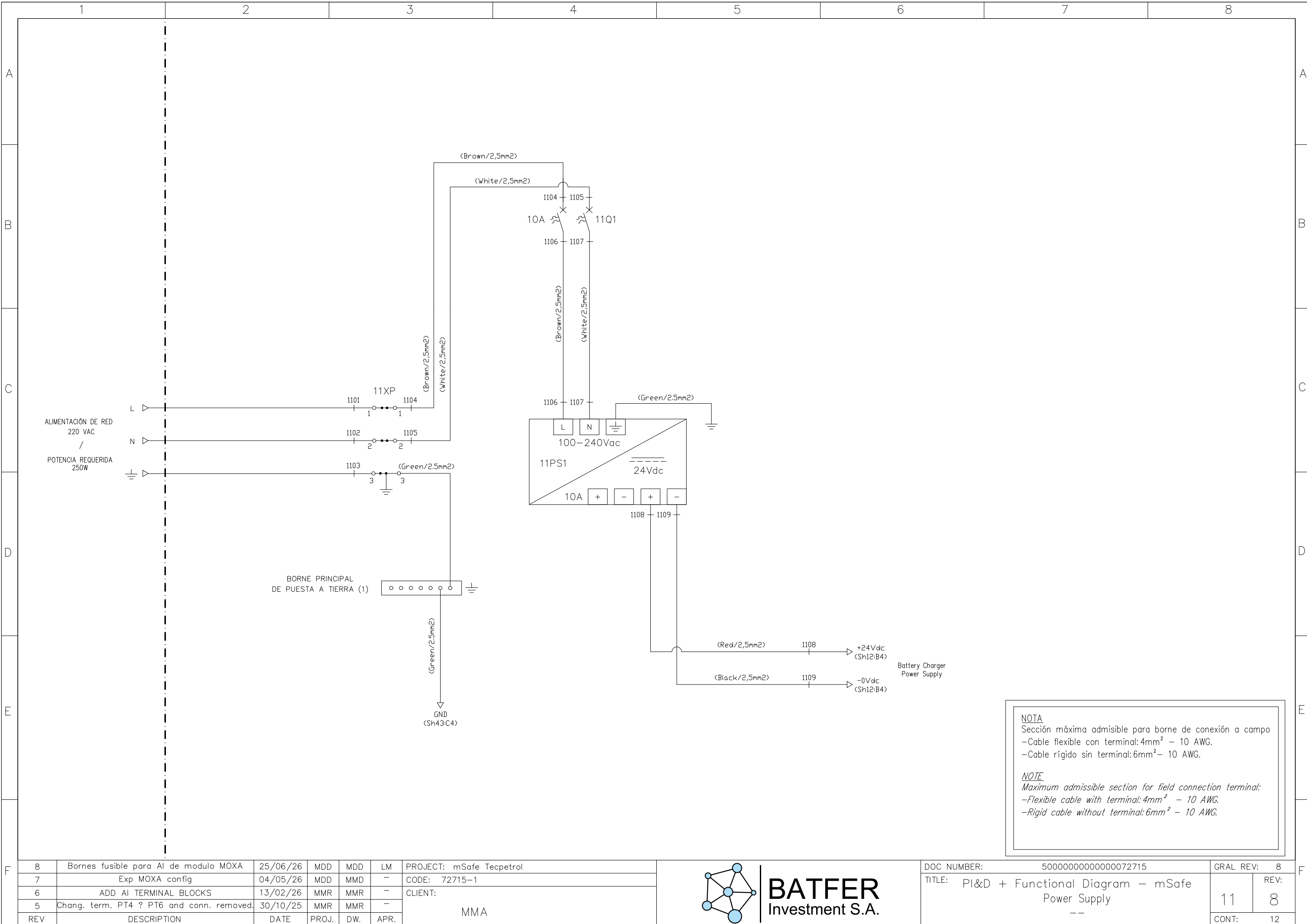
BATFER
Investment S.A.

DOC NUMBER:	50000000000000072715
TITLE:	PI&D + Functional Diagram - mSafe Pipe & Instrument Diagram P&ID

GRAL REV:	8
REV:	8
CONT:	09

1		2		3		4		5		6		7		8			
TAG		DESCRIPTION						BRAND				MODEL / CODE					
A	11XP	Power Supply Terminals / Bornes de alimentación						PHOENIX CONTACT				PT 4					
	11PS1	Power Source 110–240 Vca to 12 Vdc / Fuente de alimentación 110–240 Vac a 12 Vcc						MEAN WELL				NDR–120–12					
	11Q1	AC Control Circuit Breaker / Interruptor térmico de control AC						ABB				SH 202 – C10					
	12PS2	Battery Charger 12 Vdc / Cargador de baterías 12 Vcc						EPEVER				Tracer 5210BP					
	12XP	12Vdc Power Distribution terminal blocks / Bornes de distribución alimentación 12Vcc						PHOENIX CONTACT				PT 4					
	BH–01–M	Hidraulic Pump Motor / Motor de bomba hidráulica						BREVINI				--					
	BH–01–ZV	Hidraulic Pump Motor Starter / Arrancador motor de bomba hidráulica						BREVINI				--					
	12PB1	Power Battery / Batería de potencia						--				--					
B	12F1	Power Battery Charger Fuse / Base de fusible para cargador de baterías						ABB				E 91/32					
	13XC1	12Vdc Control Voltage Distribution / Distribución de voltaje de control 12Vcc						PHOENIX CONTACT				PT 4					
	13PS2	12/24Vdc Converter / Conversor 12/24Vcc						MEAN WELL				DDR–120A–24					
	13F2	12Vdc Control Voltage Fuse Base / Base de fusible para voltage de control 12Vcc						ABB				E 91/32					
	21PCB01	Principal Control Electronic board / Placa Electrónica de Control Principal						MMA				L03–60590–01					
	31XAI	Inner Analog Inputs Terminals / Bornes de entradas analógicas internas						PHOENIX CONTACT				PTT 2,5–2MT					
	PT–001	Hidraulic Pump Pressure Transmitter / Transmisor de presión de la bomba hidráulica						GEFRAN				KS					
	32XAI	Analog Inputs Terminals / Bornes de entradas analógicas						PHOENIX CONTACT				PTT 2,5–2MT					
C	33XAI	Analog Inputs Terminals / Bornes de entradas analógicas						PHOENIX CONTACT				PTT 2,5–2MT					
	33MX01	Analog Inputs Module / Modulo de entradas analógicas						MOXA				ioLogik R1240–T					
	43XDI	External Comand 24Vdc Terminals / Bornes de comando externo 24 Vcc						PHOENIX CONTACT				PTT 2,5–2MT					
	43DIB1	Digital Input signals – Intrinsic safety barrier / Entrada Digital – Barrera Seg. Intrínseca						CHENZHU				GS8512–EX.22					
	43XDIB	Intrinsic Safety Digital Inputs Terminals / Bornes de entradas digitales Seg. Intrínseca						PHOENIX CONTACT				PTT 2,5–2 MT BU					
	43KR	Terminal Relay 24Vdc / Borne relé 24Vcc						PHOENIX CONTACT				RIF–0–RPT–24DC/21					
	61KR1..3	Terminal Relay / Borne relé						PHOENIX CONTACT				RIF–0–RPT–12DC/21					
	XM1	Commands Terminals / Bornes de comandos						PHOENIX CONTACT				PTT 2,5–2MT					
D	ZY	Solenoid Valve / Válvula solenoide						BREVINI				CRP04 NA 18W					
	81XCM	Communication Terminal / Borne de comunicación						PHOENIX CONTACT				PTT 2,5–2MT					
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8		Bornes fusible para AI de modulo MOXA		25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol				DOC NUMBER: 5000000000000072715				GRAL REV: 8	
7		Exp MOXA config		04/05/26	MDD	MMD	–	CODE: 72715–1				TITLE: PI&D + Functional Diagram – mSafe				REV: 8	
6		ADD AI TERMINAL BLOCKS		13/02/26	MMR	MMR	–	CLIENT: MMA				Material's List				09	
5		Chang. term. PT4 ? PT6 and conn. removed.		30/10/25	MMR	MMR	–										
REV		DESCRIPTION		DATE	PROJ.	DW.	APR.									CONT: 11	

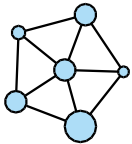
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NOTA
Sección máxima admisible para borne de conexión a campo
-Cable flexible con terminal:4mm² - 10 AWG.
-Cable rígido sin terminal:6mm²- 10 AWG.

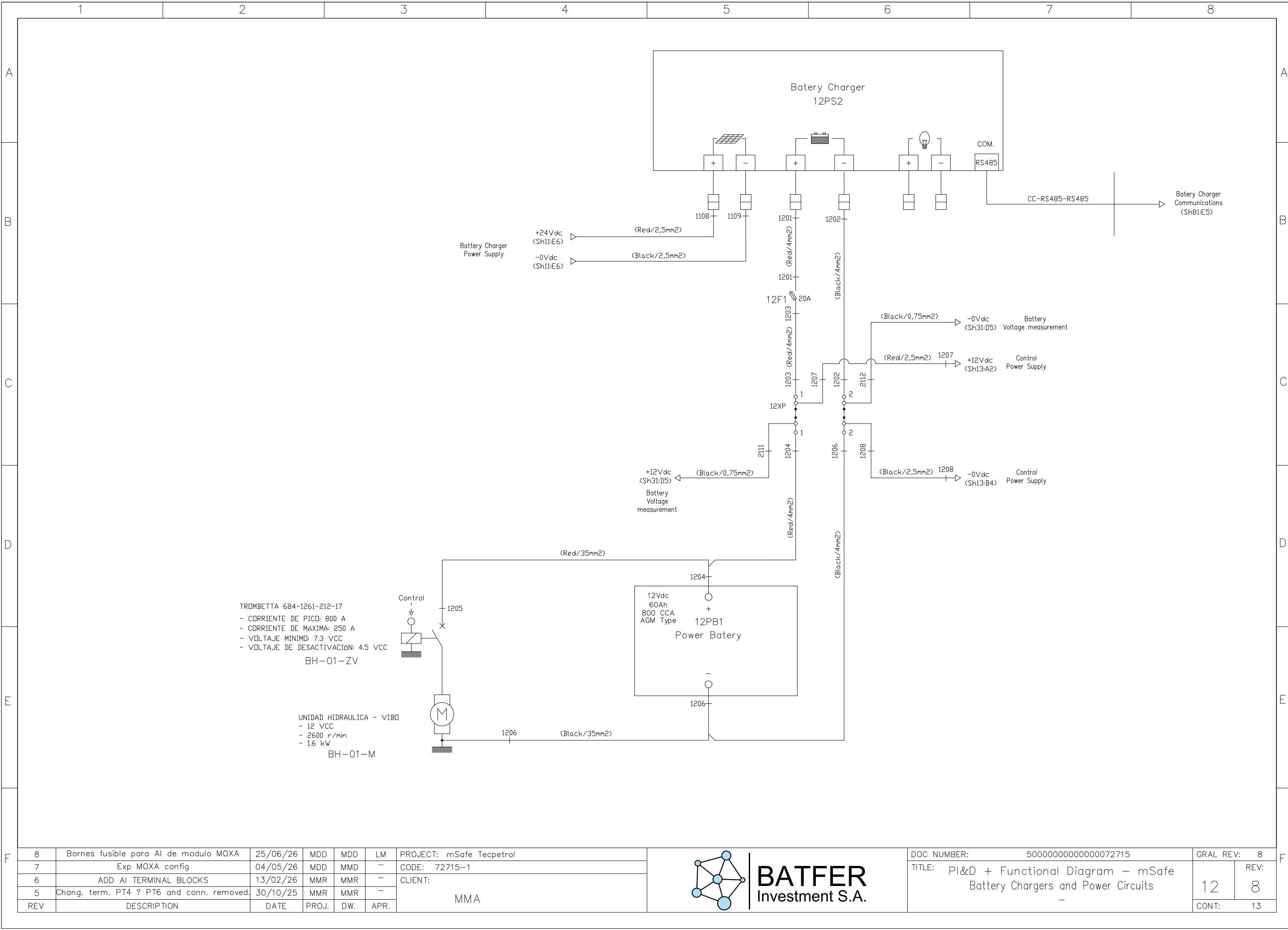
NOTE
Maximum admissible section for field connection terminal:
-Flexible cable with terminal:4mm² - 10 AWG.
-Rigid cable without terminal:6mm² - 10 AWG.

8	Bornes fusible para AI de modulo MOXA	25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol
7	Exp MOXA config	04/05/26	MDD	MMD	-	CODE: 72715-1
6	ADD AI TERMINAL BLOCKS	13/02/26	MMR	MMR	-	CLIENT:
5	Chang. term. PT4 ? PT6 and conn. removed.	30/10/25	MMR	MMR	-	
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.	MMA



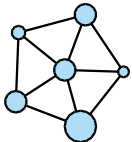
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DOC NUMBER:	5000000000000072715	GRAL REV:	8
TITLE:	PI&D + Functional Diagram - mSafe Power Supply		REV:
	--	11	8
		CONT:	12



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8	Bornes fusible para AI de modulo MOXA	25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol
7	Exp MOXA config	04/05/26	MDD	MMD	-	CODE: 72715-1
6	ADD AI TERMINAL BLOCKS	13/02/26	MMR	MMR	-	CLIENT:
5	Chang. term. PT4 ? PT6 and conn. removed.	30/10/25	MMR	MMR	-	
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.	MMA

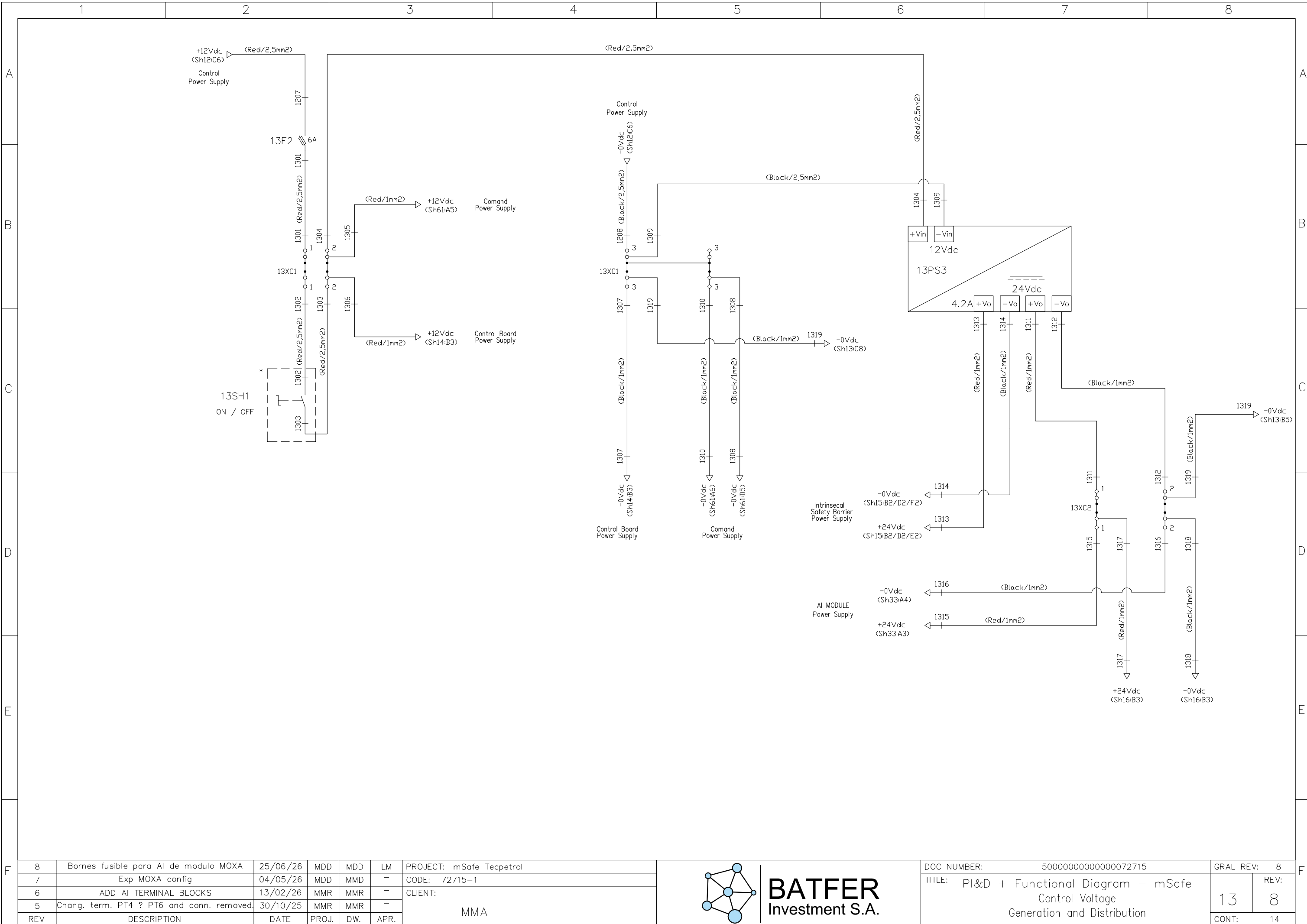


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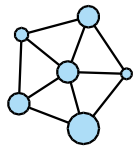
DOC NUMBER:	5000000000000072715	GRAL REV:	8
TITLE:	PI&D + Functional Diagram - mSafe Battery Chargers and Power Circuits		
	-		
CONT:	13		

REV:	8
CONT:	13

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8	Bornes fusible para AI de modulo MOXA	25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol
7	Exp MOXA config	04/05/26	MDD	MMD	-	CODE: 72715-1
6	ADD AI TERMINAL BLOCKS	13/02/26	MMR	MMR	-	CLIENT:
5	Chang. term. PT4 ? PT6 and conn. removed.	30/10/25	MMR	MMR	-	
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.	MMA

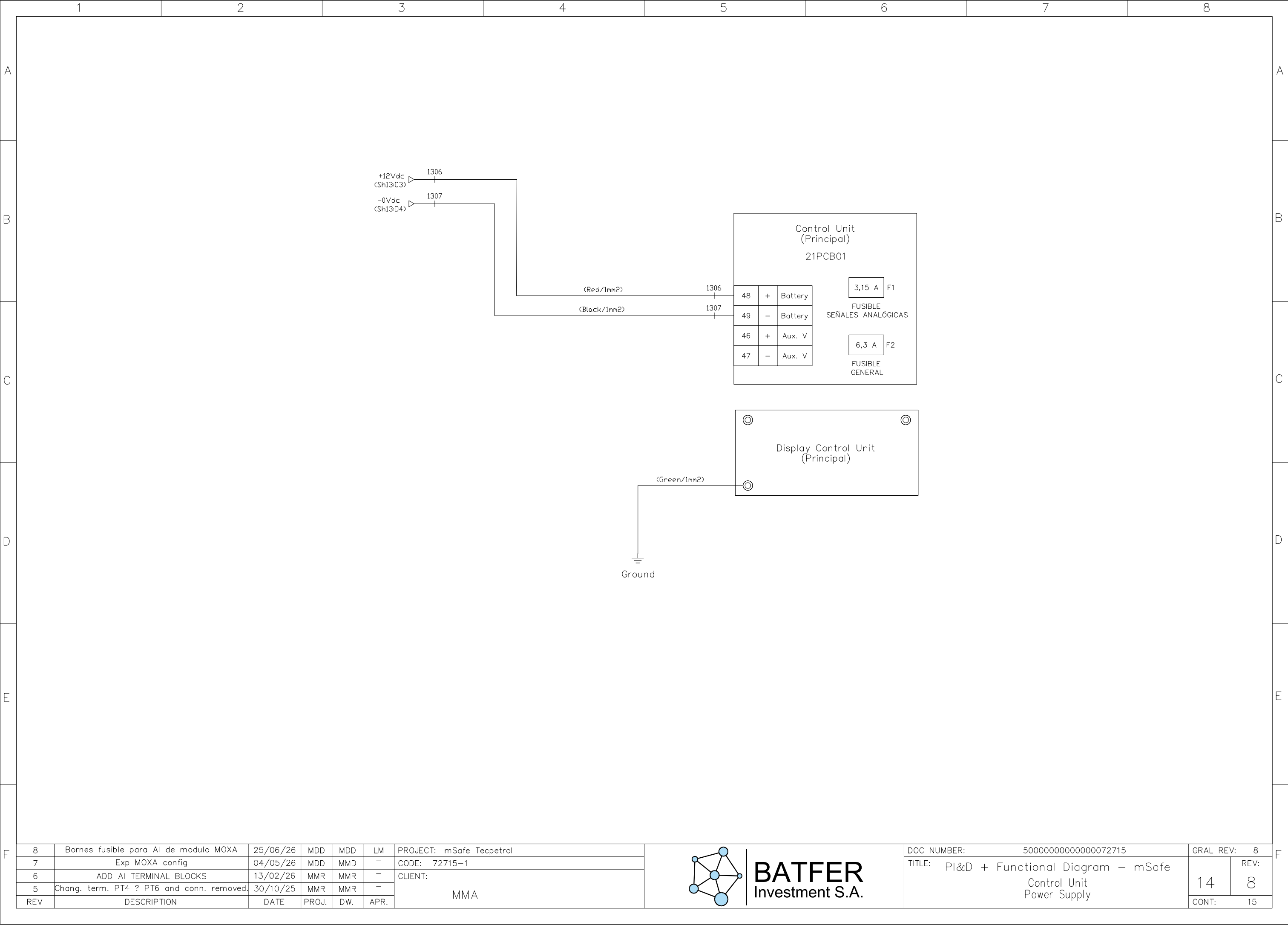


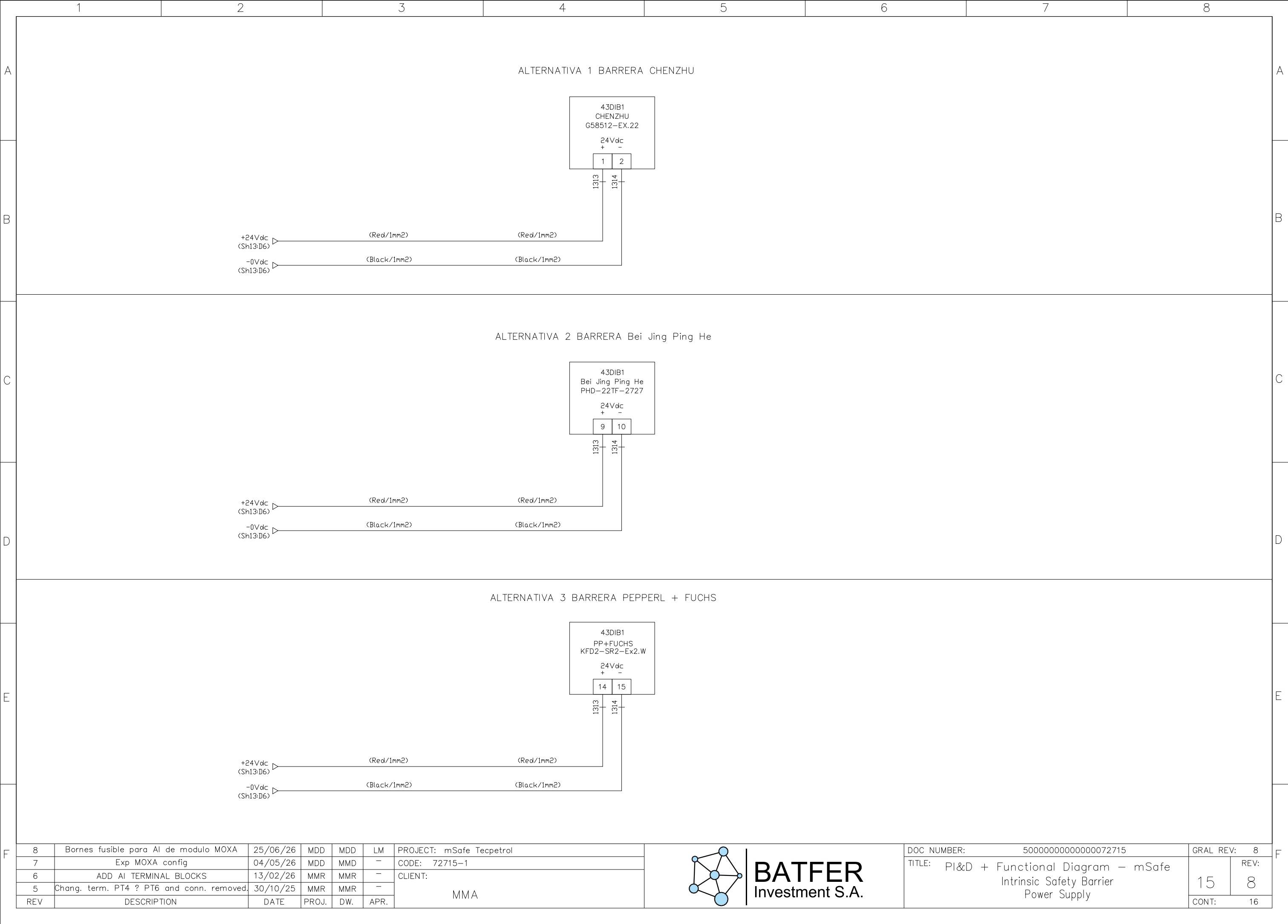
BATFER
Investment S.A.

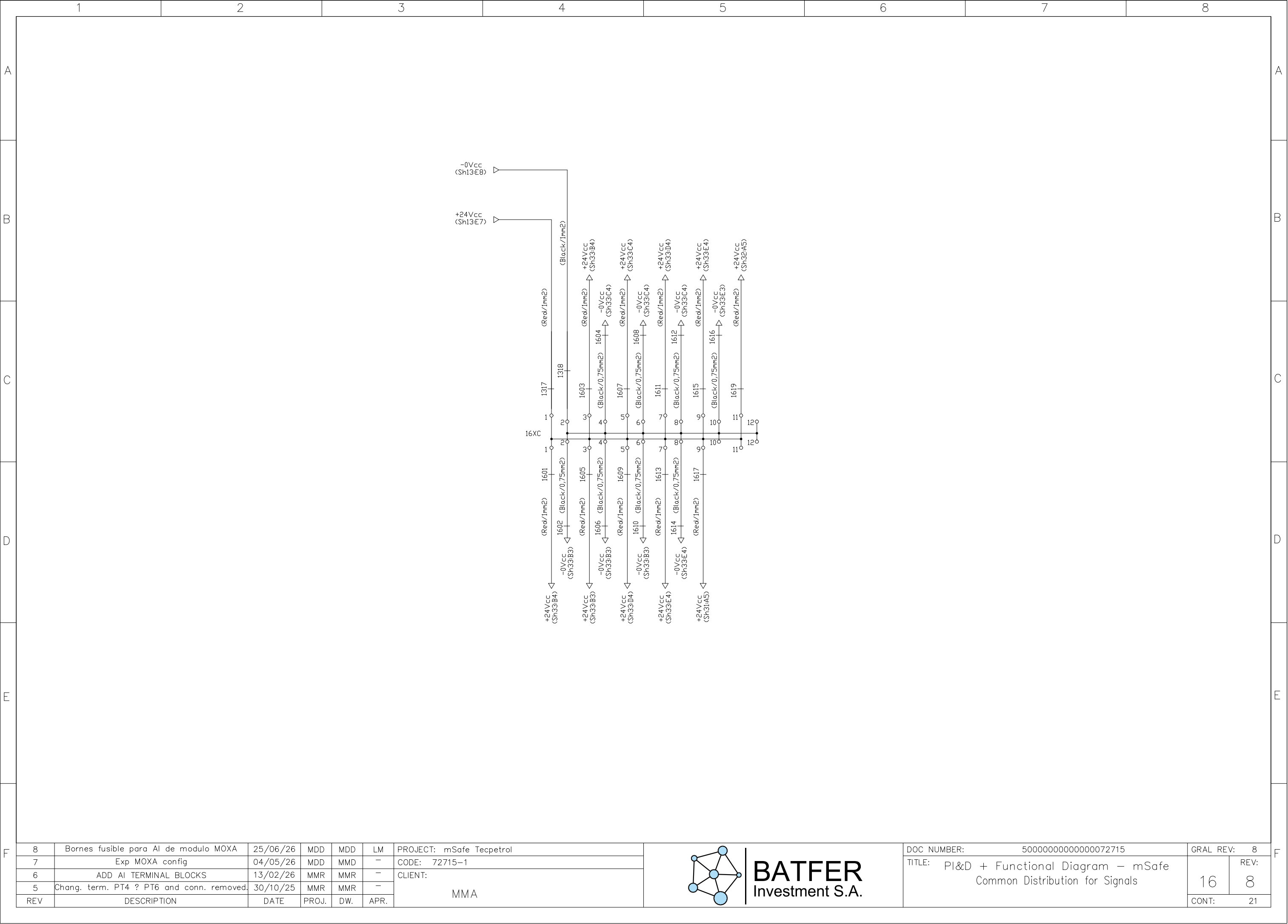
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TITLE:	PI&D + Functional Diagram - mSafe Control Voltage Generation and Distribution

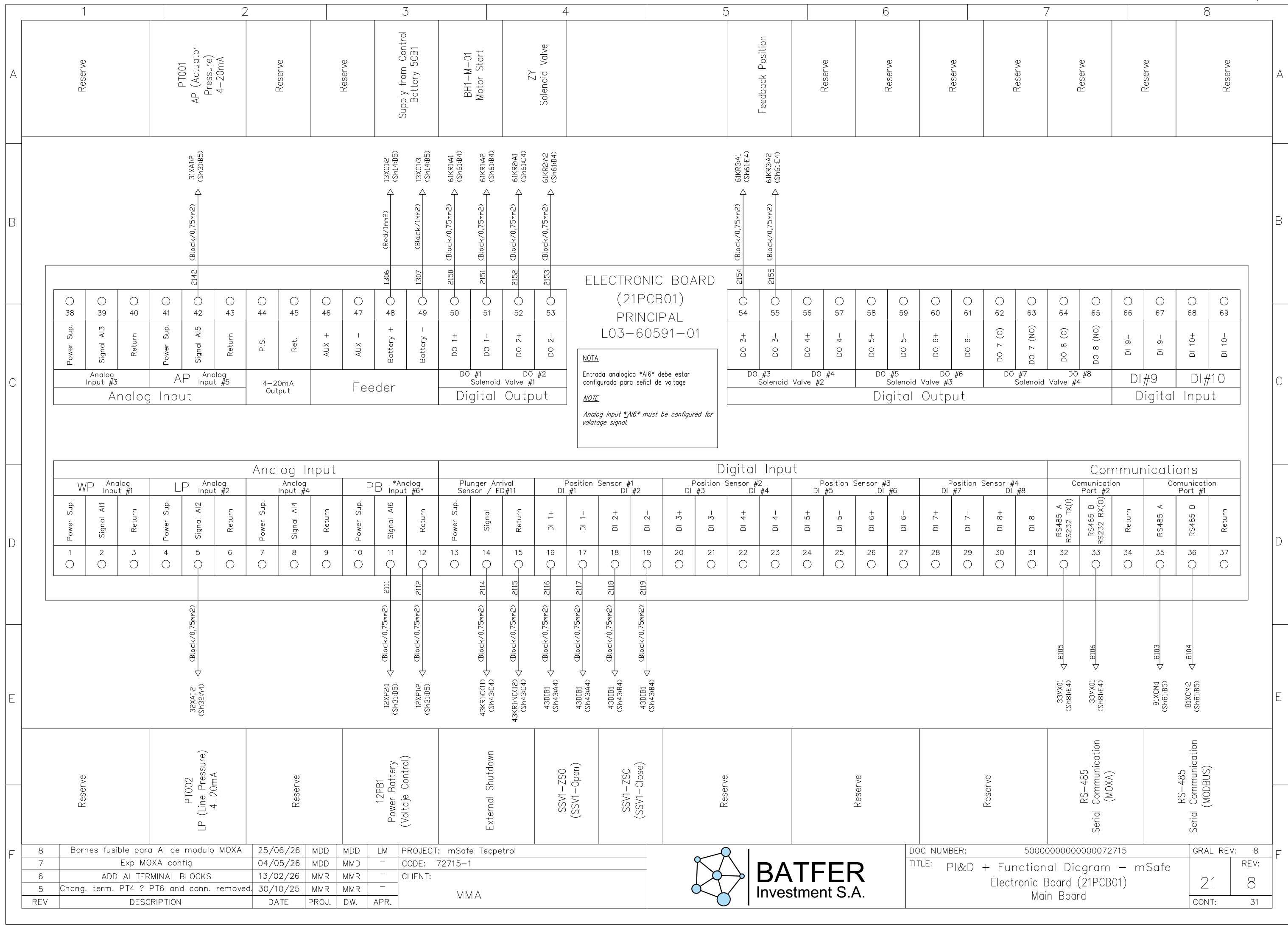
GRAL REV:	8
REV:	8
CONT:	14

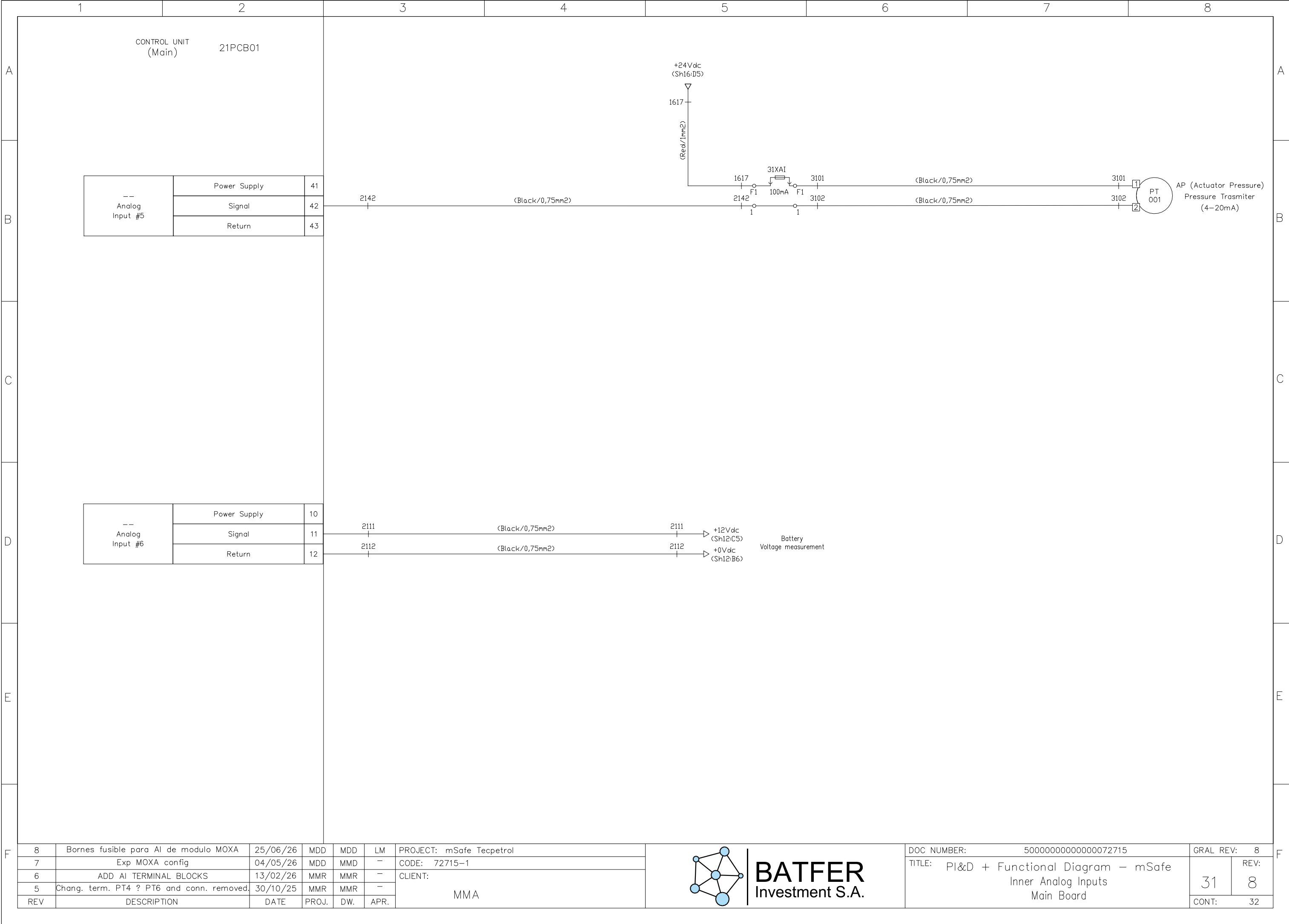
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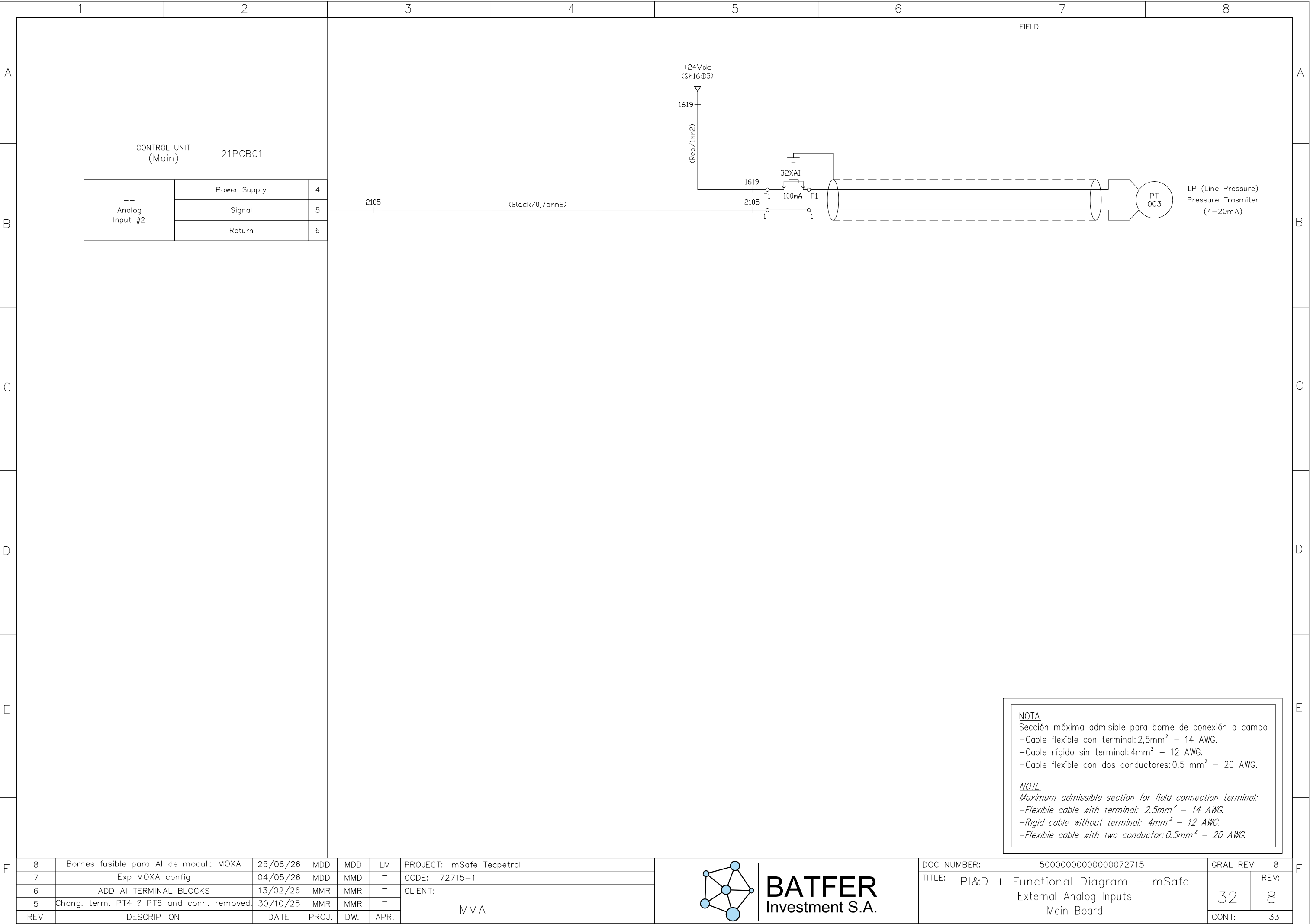






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NOTE: cables not indicated will be black 0.75mm²/18 AWG



NOTA

Sección máxima admisible para borne de conexión a campo

-Cable flexible con terminal:2,5mm² - 14 AWG.

-Cable rígido sin terminal:4mm² - 12 AWG.

-Cable flexible con dos conductores:0,5 mm² - 20 AWG.

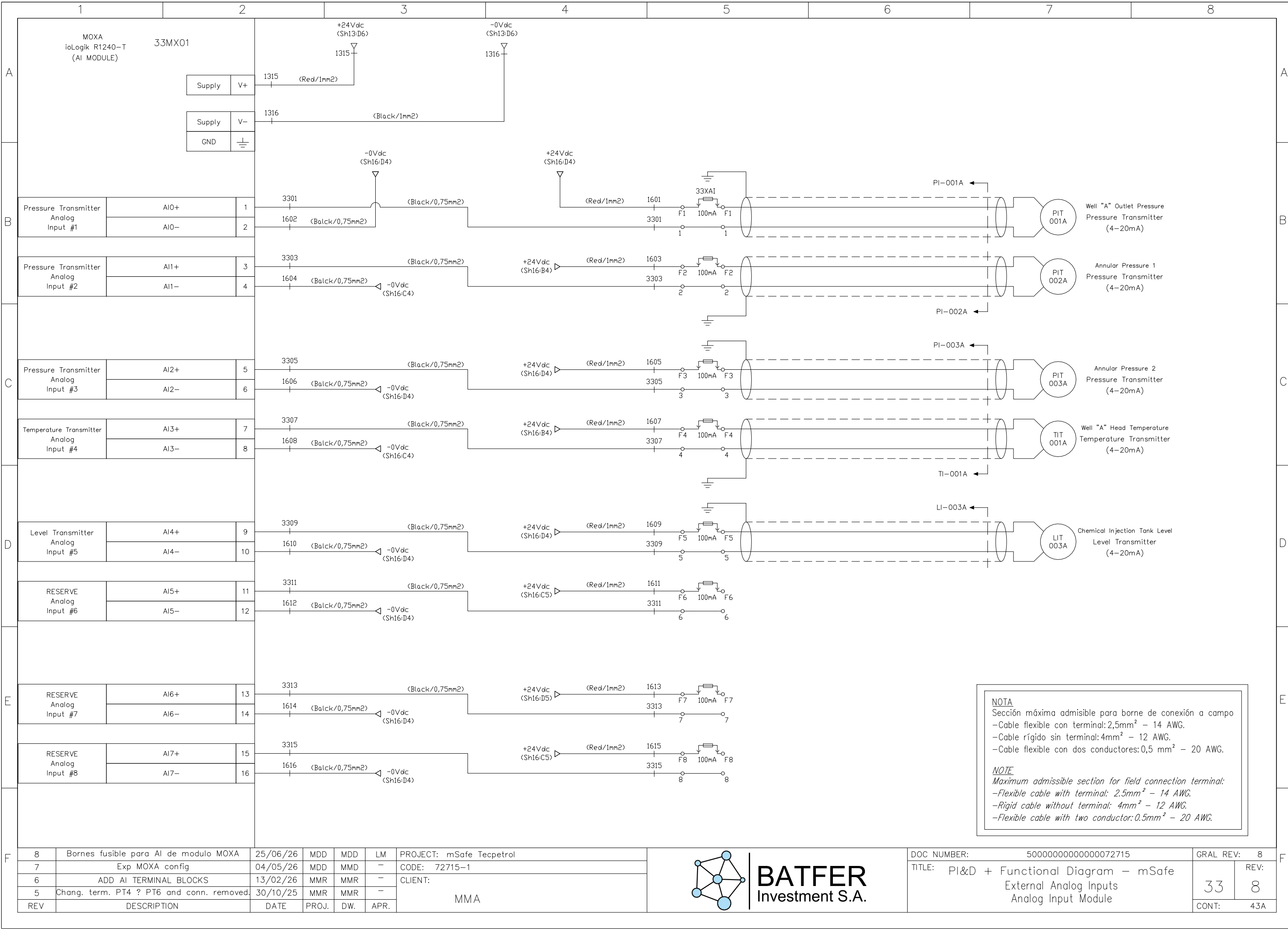
NOTE

Maximum admissible section for field connection terminal:

-Flexible cable with terminal: 2.5mm² - 14 AWG.

-Rigid cable without terminal: 4mm² - 12 AWG.

-Flexible cable with two conductor:0.5mm² - 20 AWG.



NOTE

Sección máxima admisible para borne de conexión a campo

-Cable flexible con terminal: 2,5mm² - 14 AWG.

-Cable rígido sin terminal: 4mm² - 12 AWG.

-Cable flexible con dos conductores: 0,5 mm² - 20 AWG.

NOTE

Maximum admissible section for field connection terminal:

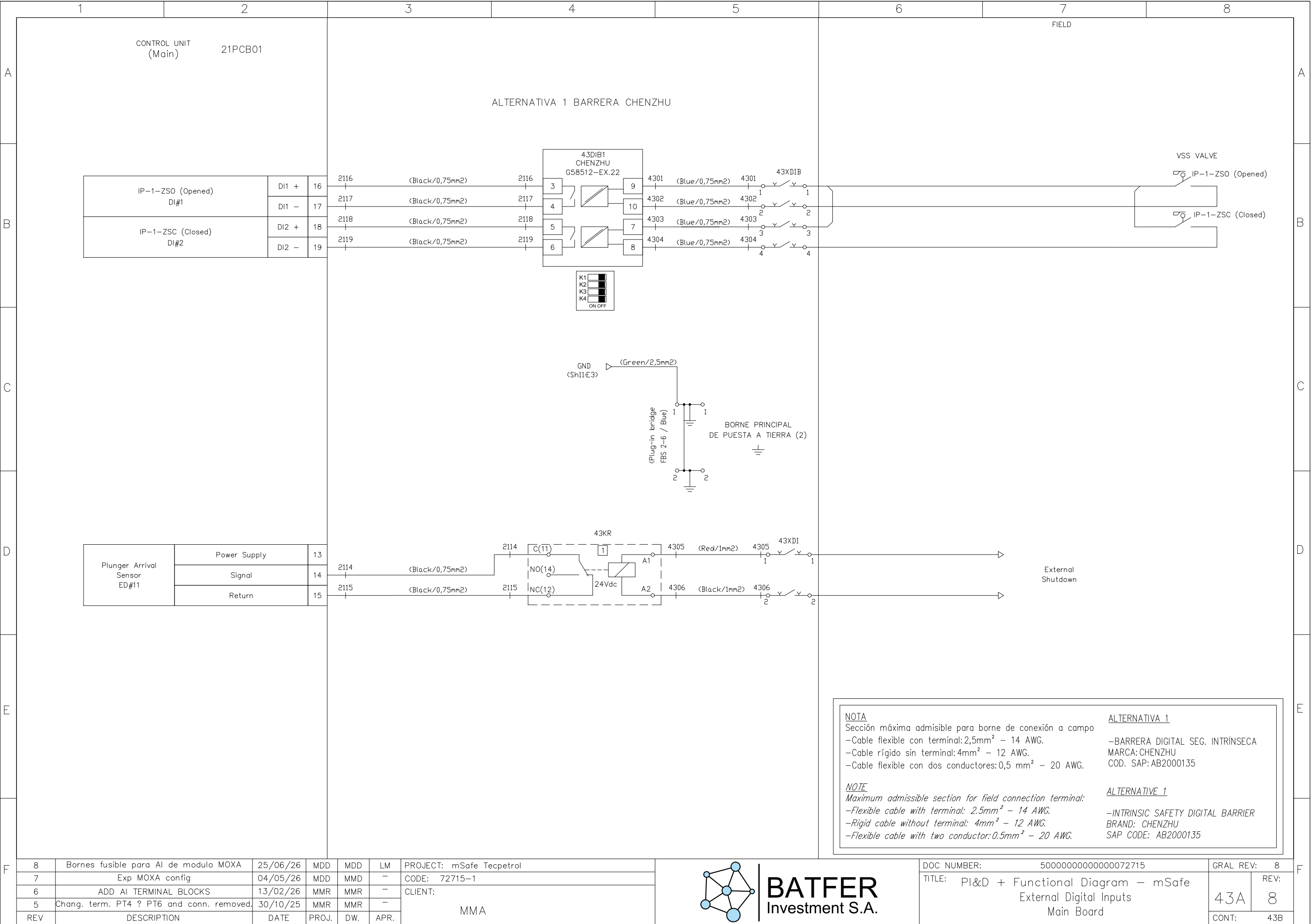
-Flexible cable with terminal: 2.5mm² - 14 AWG.

-Rigid cable without terminal: 4mm² - 12 AWG.

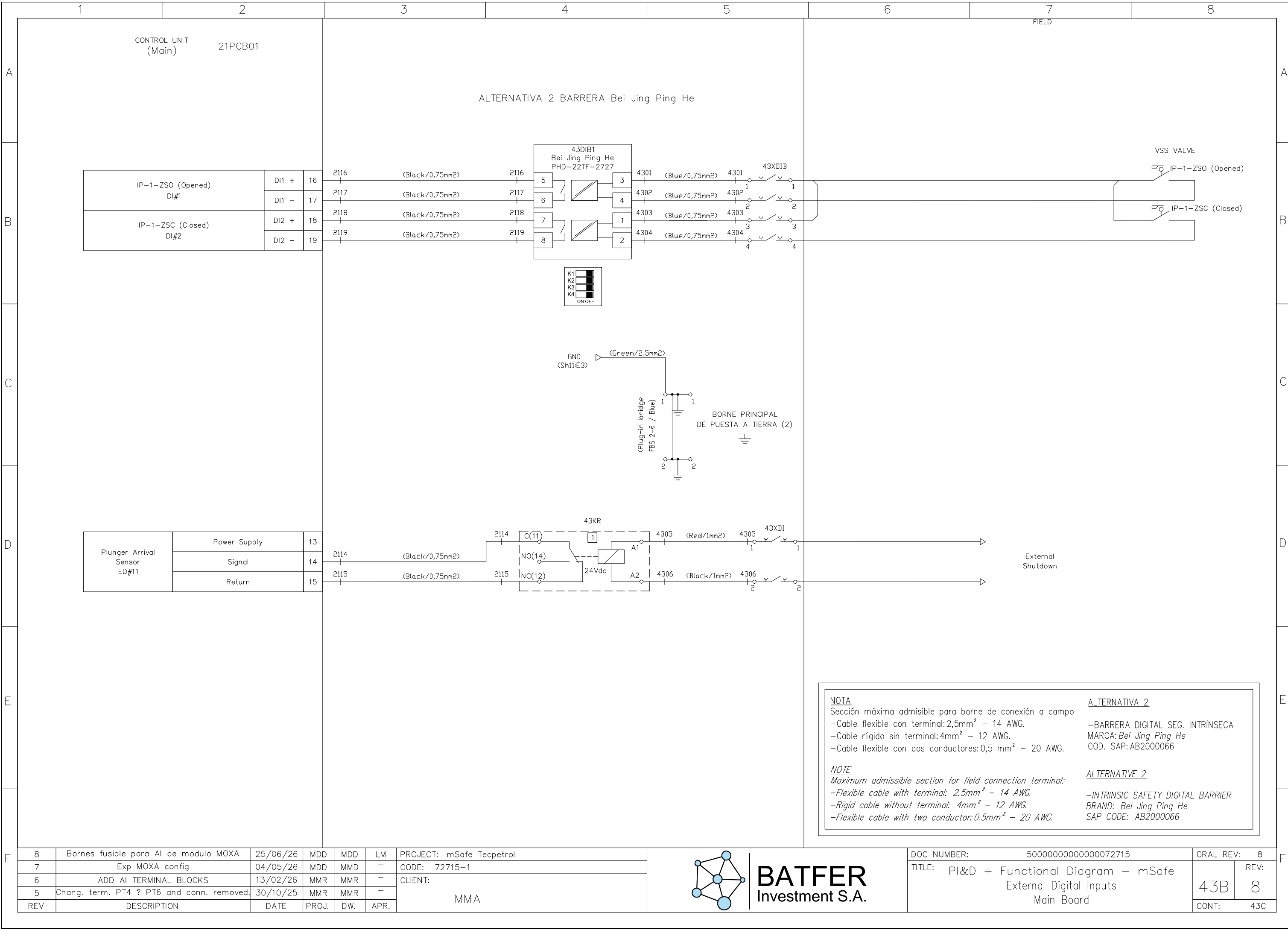
-Flexible cable with two conductor: 0.5mm² - 20 AWG.

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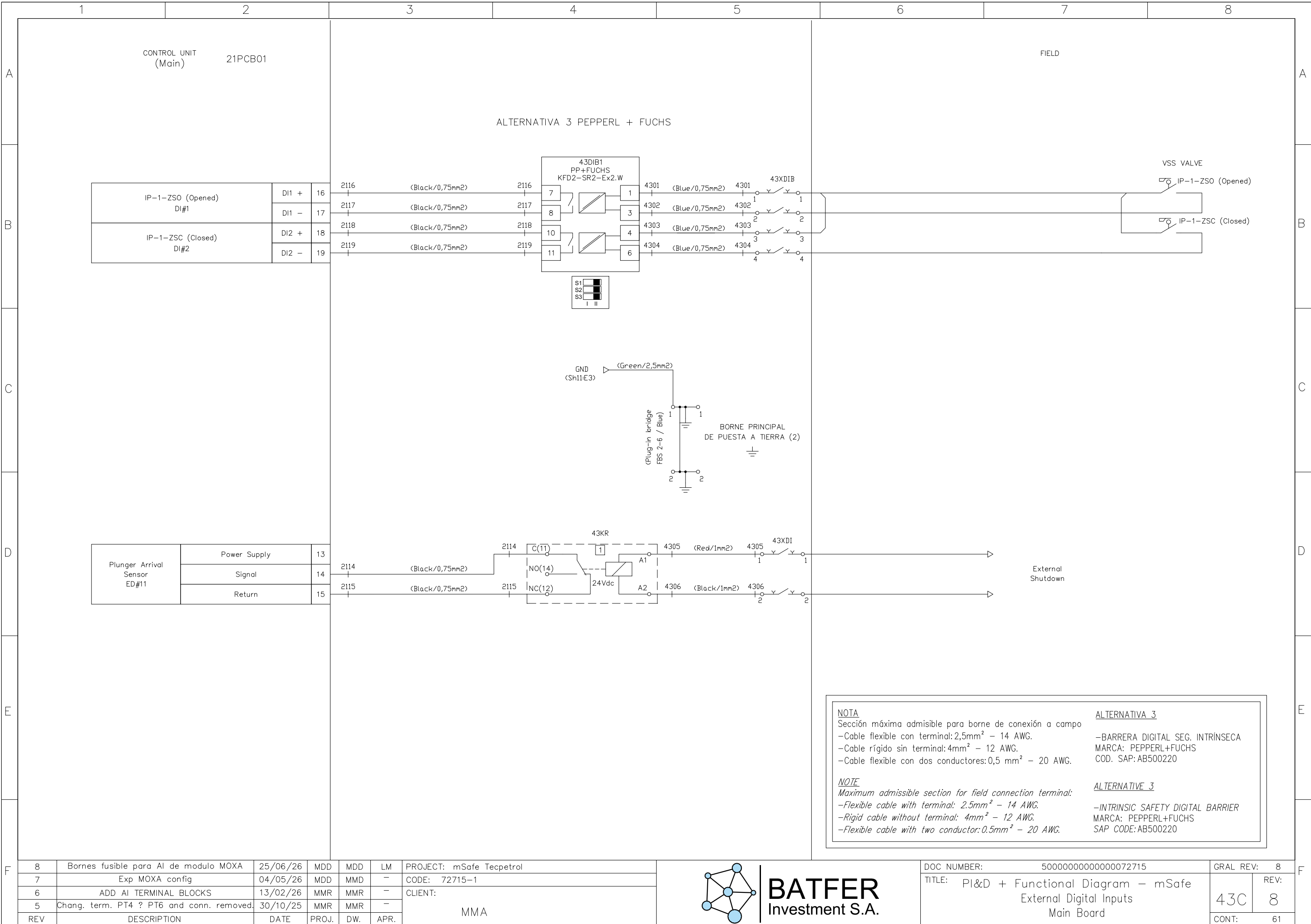
NOTE: cables not indicated will be black 0.75mm²/18 AWG



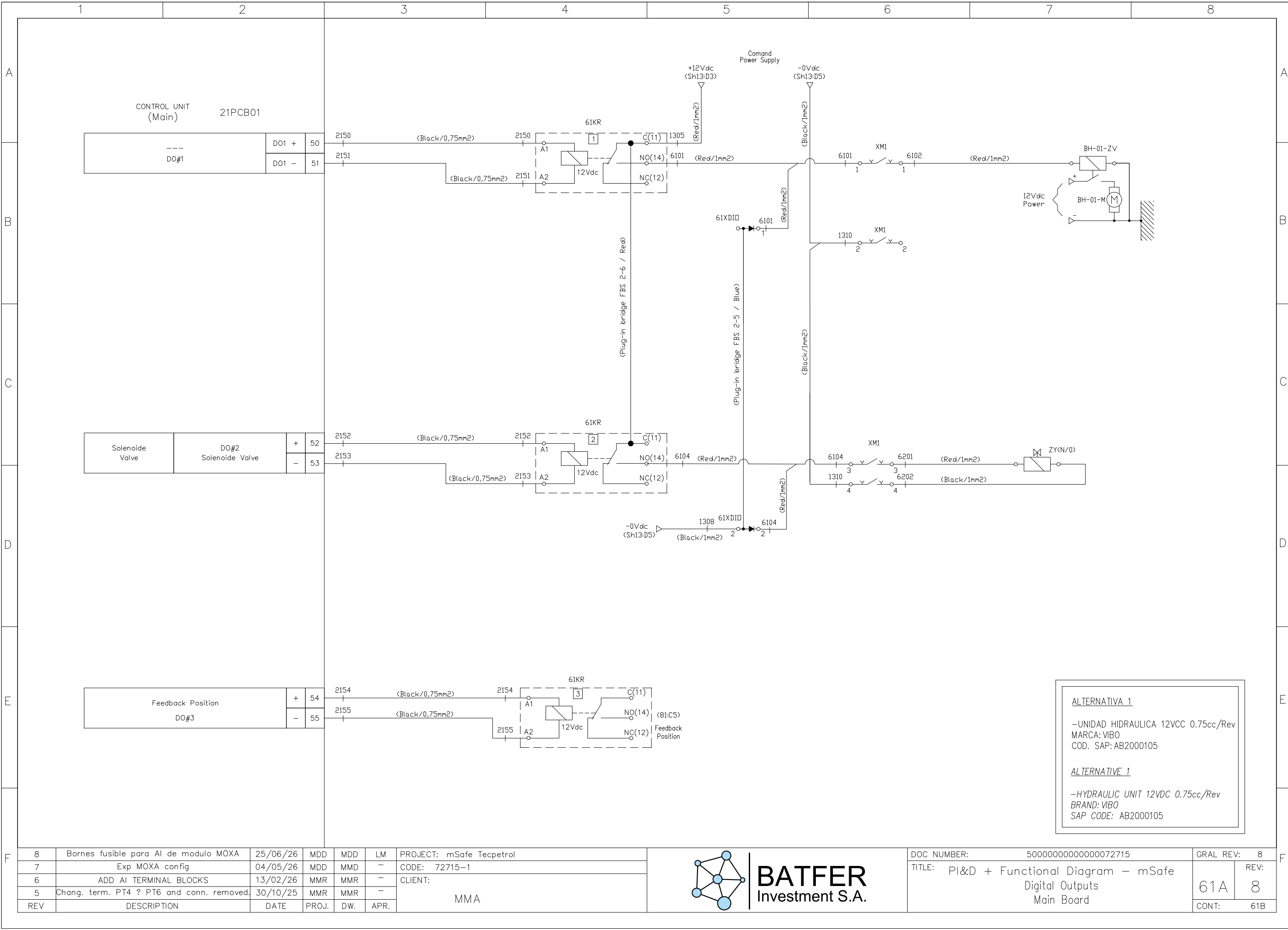
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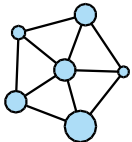
ALTERNATIVA 1

–UNIDAD HIDRAULICA 12VCC 0.75cc/Rev
MARCA: VIBO
COD. SAP: AB2000105

ALTERNATIVE 1

–HYDRAULIC UNIT 12VDC 0.75cc/Rev
BRAND: VIBO
SAP CODE: AB2000105

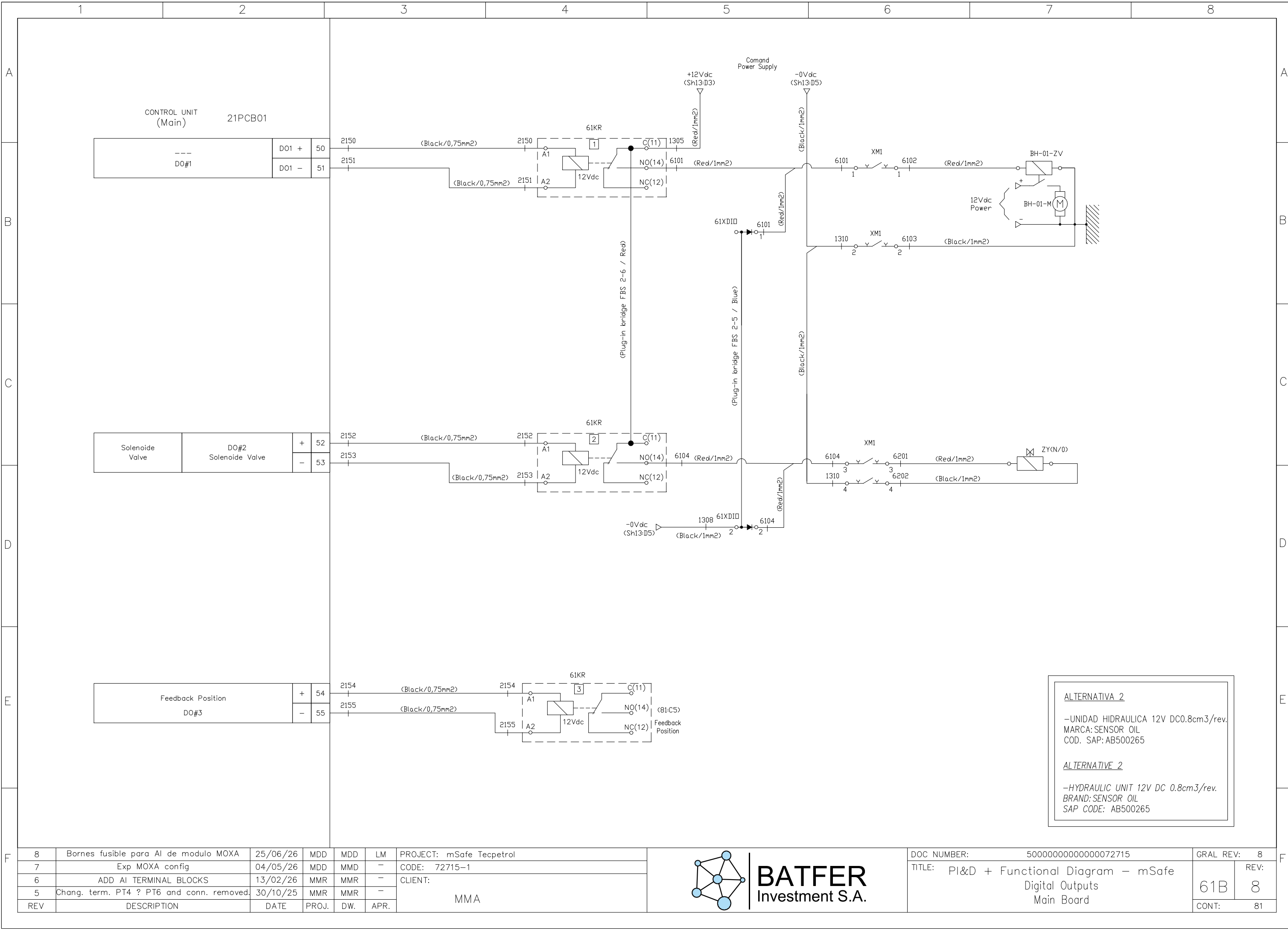
8	Bornes fusible para AI de modulo MOXA	25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol
7	Exp MOXA config	04/05/26	MDD	MMD	–	CODE: 72715–1
6	ADD AI TERMINAL BLOCKS	13/02/26	MMR	MMR	–	CLIENT:
5	Chang. term. PT4 ? PT6 and conn. removed.	30/10/25	MMR	MMR	–	
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.	MMA



BATFER
Investment S.A.

DOC NUMBER:	5000000000000072715	GRAL REV:	8
TITLE:	PI&D + Functional Diagram – mSafe Digital Outputs Main Board	REV:	8
		CONT:	61B

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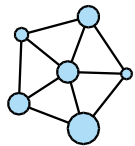
ALTERNATIVA 2

—UNIDAD HIDRAULICA 12V DC 0.8cm³/rev.
MARCA: SENSOR OIL
COD. SAP: AB500265

ALTERNATIVE 2

—HYDRAULIC UNIT 12V DC 0.8cm³/rev.
BRAND: SENSOR OIL
SAP CODE: AB500265

8	Bornes fusible para AI de modulo MOXA	25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol
7	Exp MOXA config	04/05/26	MDD	MMD	—	CODE: 72715—1
6	ADD AI TERMINAL BLOCKS	13/02/26	MMR	MMR	—	CLIENT:
5	Chang. term. PT4 ? PT6 and conn. removed.	30/10/25	MMR	MMR	—	
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.	MMA



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DOC NUMBER:	5000000000000072715
TITLE:	PI&D + Functional Diagram — mSafe Digital Outputs Main Board

GRAL REV:	8
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